

A Reduced BM Four-Frame from Time, Space, Energy and Momentum

A Spectral-Geometric Perspective on Prime Structures

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Abstract

We introduce a reduced four-dimensional framework of the Bamberg Model (BM) consisting of four distinguished directions (A, C, e, b) interpreted as time, space, energy and momentum. Within this setting a Minkowski-type structure arises naturally.

We show that this four-frame admits a Clifford structure and therefore defines a Dirac-type operator

$$D_{\text{BM}} = \Gamma_A A + \Gamma_C C + \Gamma_e e + \Gamma_b b.$$

The square of this operator reproduces the invariant

$$D_{\text{BM}}^2 = A^2 - C^2 + e^2 - b^2.$$

Motivated by the Hilbert–Pólya idea we explore the possibility that the imaginary parts of the nontrivial Riemann zeros may appear as spectral values of such an operator.

The work is presented as a heuristic program connecting arithmetical structures of primes with spectral geometry.

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1 Introduction

The distribution of prime numbers and the spectral properties of the Riemann zeta function are among the deepest open problems of mathematics.

The Hilbert–Pólya conjecture proposes that the imaginary parts of the nontrivial zeros

$$\rho_n = \frac{1}{2} + i\gamma_n$$

arise as eigenvalues of a self-adjoint operator.

At the same time spectral geometry has shown that geometric structures can be encoded in the spectrum of Dirac operators.

The present work proposes a reduced geometric framework, referred to as the Bamberg Model (BM), based on four directions

$$(A, C, e, b)$$

interpreted as time, space, energy and momentum.

The goal is not to prove the Riemann hypothesis but to formulate a consistent spectral-geometric setting linking arithmetical prime structures with Dirac-type operators.

2 The BM Four-Frame

We introduce four distinguished directions

$$(A, C, e, b).$$

They are interpreted as

Symbol	Interpretation
A	time direction
C	spatial direction
e	energy coordinate
b	momentum coordinate

The first pair forms a reduced space-time structure, while the second pair corresponds to energy-momentum variables.

3 Reduced Space-Time Structure

Assuming A to be time-like and C space-like, a Minkowski-type interval arises naturally:

$$ds^2 = c_M^2 dA^2 - dC^2.$$

This defines an effective light cone

$$|v| = \left| \frac{dC}{dA} \right| \leq c_M.$$

4 Energy–Momentum Structure

The energy-momentum analogue of the Minkowski interval is

$$e^2 - c_M^2 b^2.$$

This leads to a relativistic invariant

$$e^2 - c_M^2 b^2 = m_{\text{BM}}^2 c_M^4.$$

5 Clifford Structure

The four BM directions naturally define a Clifford algebra

$$Cl(1, 3)$$

with generators Γ_μ satisfying

$$\{\Gamma_\mu, \Gamma_\nu\} = 2g_{\mu\nu}.$$

6 BM Dirac Operator

We define a Dirac-type operator

$$D_{\text{BM}} = \Gamma_A A + \Gamma_C C + \Gamma_e e + \Gamma_b b.$$

Using the Clifford relations one obtains

$$D_{\text{BM}}^2 = A^2 - C^2 + e^2 - b^2.$$

In the light-like case

$$A^2 - C^2 = 0$$

this reduces to

$$D_{\text{BM}}^2 = e^2 - b^2.$$

7 Prime Structure

The multiplicative group

$$(\mathbb{Z}/12\mathbb{Z})^\times = \{1, 5, 7, 11\}$$

forms the Klein four group

$$V_4.$$

Prime quadruples of these residue classes appear naturally in the BM framework.

8 Numerical Experiments

To explore the BM framework numerically we consider prime quadruples

$$(p_1, p_2, p_3, p_4)$$

with residues

$$\{1, 5, 7, 11\} \pmod{12}.$$

Coordinates are defined as

$$C = \log(p_1 p_2 p_3 p_4)$$

and

$$b = \log(p_1^2 + p_2^2 + p_3^2 + p_4^2).$$

Combined with Riemann zero ordinates γ_n we obtain BM coordinates

$$(A, C, e, b).$$

9 BM Dirac Spectrum

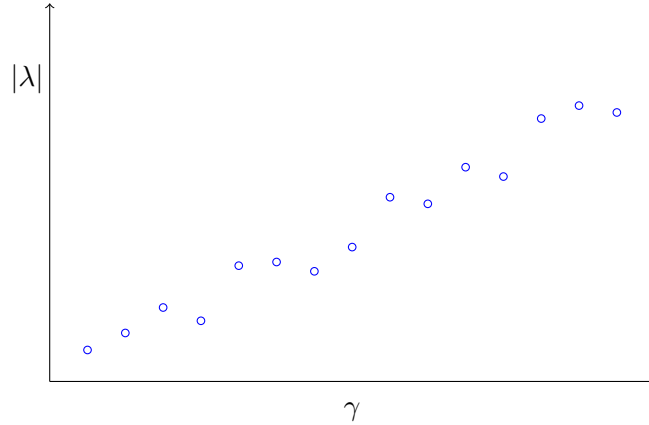


Figure 1: Illustration of Dirac-type spectral scaling $\lambda \sim \gamma$.

10 Prime Shell Structure

11 Discussion

The BM construction should be understood as a heuristic framework.

The numerical experiments do not establish a direct spectral identity between the BM Dirac operator and the Riemann zeros.

However the emergence of Dirac-like spectral structures suggests that the four-frame (A, C, e, b) may provide a useful geometric interpretation of arithmetic patterns.

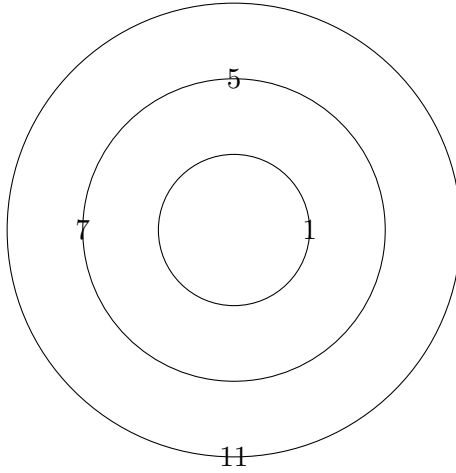


Figure 2: Schematic illustration of V4 prime shells in the BM framework.

12 Outlook

Future work may include

- construction of a self-adjoint BM Dirac operator
- deeper analysis of the spectral distribution
- connections to noncommutative spectral geometry

References

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